Furnace Analysis

Know how worn the lining of every steel converter is, heat by heat, and get warned before it becomes unsafe.

WHAT IT DOES

The converter's inside is lined with bricks that wear thin. After a heat, the vendor scans the lining and sends a PDF report. Furnace Analysis reads that report, checks every part of the lining against safe limits, and shows what needs attention. No more hunting through PDFs and spreadsheets.

HOW IT WORKS

1 Upload
The analyst drops in the vendor's PDF for one heat.

2 Read
The app reads the 22 lining zones for you (about 30 seconds).

3 Check & save
The analyst compares with the PDF, fixes any slip, then saves.

4 Alert
Every zone gets a colour. If any is Warning or Critical, an email goes out.

THE COLOURS

OK — lining is thick enough
Warning — getting thin, keep watching
Critical — too thin, act now
No limit — no limit set for that zone yet

Each zone has two numbers in mm: **Average** and **Minimum** thickness. The worse of the two decides the colour.

Limits are set by an administrator for each converter: *Warning below* and *Critical below* (in mm), for one zone, a whole section or all zones. Every change is saved as a new version, so history is kept. Each limit can carry a recommendation, for example "Stop and inspect".

WHAT YOU SEE

Dashboard	Latest scan, count of Critical / Warning / OK zones, the 8 zones needing attention most, and recent scans.
Scan page	All 22 zones in four sections (Tapside, Trunnion, Charge Pad, Slag line) with colour, change since the previous scan, and a **Repair?** flag when the lining got thicker by 30 mm or more (it cannot thicken by wear, so it was probably patched). Opens the vendor's original PDF.
Trends	A chart of one zone's Average and Minimum over all scans, with the warning and critical lines drawn on it.
Settings	Administrators manage converters, limits (type them or upload Excel), email recipients and who may upload.

WHO CAN DO WHAT

Viewer	Analyst	Administrator
See dashboard, scans, trends and PDFs	Everything a Viewer can, plus upload and save reports	Everything an Analyst can, plus Settings (limits, converters, recipients, roles)

Sign in with User ID and password. Each person only sees the applications they have been given. Times are shown in Indian time (IST).

EXAMPLE

Heat **2886** (measurement 25, scanned 1 Nov 2026) of Converter 1, judged with Warning below 400 mm and Critical below 150 mm: **2 zones Critical** (Right Trunion-1 at 108 mm and ChargePad-1 at 130 mm), **7 Warning**, **13 OK**. The dashboard lists them worst first and an email goes to the recipients.

NOT INCLUDED

Offline use, editing or deleting a saved scan, generated PDF reports (the vendor's PDF is kept and can be opened), in-app notifications (email is used).